06-01 第二篇

Cooking and Chemistry: Science in the Kitchen

When you step into a kitchen, you might not think about science. But every time you bake a cake, boil an egg, or fry a piece of meat, you are actually doing chemistry. [1] Cooking is a wonderful way to see science in action, and understanding it can help you become a better cook. Many of the changes we see in food—browning, rising, and melting—are chemical reactions happening right before our eyes.

Take bread for example. When you add yeast（酵母） to flour and water, something amazing happens. [2] The yeast is a tiny living organism that eats sugar and produces carbon dioxide gas. This gas forms bubbles in the dough, making it rise and become soft and fluffy（松软的）. Without yeast, bread would be flat and hard. The same principle works when you make yogurt or cheese—bacteria（细菌） change the chemical structure of milk.

Why does an egg turn solid when you heat it? [3] The protein（蛋白质） molecules in the egg unfold and connect to each other when they get hot. This changes the liquid egg into a solid form. It is similar to what happens when you make jelly—gelatin（明胶） proteins form a network that holds water and creates a gel. Understanding these changes can help you know exactly when your food is perfectly cooked.

Some cooking tricks are actually chemistry lessons. [4] Adding a pinch of salt to boiling water can make pasta cook faster, because salt raises the boiling point of water slightly. Marinating（腌制） meat in lemon juice or vinegar before cooking makes it more tender, because the acid breaks down the muscle fibers. Even the way you cut vegetables can affect how they taste—smaller cuts release more flavor when cooked.

[5] Cooking and chemistry go hand in hand. The kitchen is like a lab where we can experiment, learn, and create delicious results. So the next time you cook, take a moment to think about the science behind your food—it might make your meal even more interesting.

选项

A. In other words, the kitchen is a place where we can discover how science works in everyday life. B. That is because the yeast causes a chemical reaction that changes the structure of the dough. C. In fact, the way food changes when we cook it follows clear scientific rules. D. These are practical examples of how knowing a little chemistry can improve your cooking. E. In short, every meal you make is a chance to learn something new about science. F. For instance, professional chefs often study food science to improve their skills.

答案键（Answer Key）

[1] A [2] B [3] C [4] D [5] E

深度解析

- **[1] A**：
  - **线索**：引出厨房中的科学。空前描述"every time you bake a cake, boil an egg, or fry...you are actually doing chemistry"，空后说明"Cooking is a wonderful way to see science in action...Many of the changes we see in food...are chemical reactions"。A选项"In other words, the kitchen is a place where we can discover how science works in everyday life"（换言之，厨房是一个我们发现科学在日常工作中如何运作的地方）中"In other words"概括空前烹饪即化学，"discover how science works"与空后"see science in action"呼应，衔接顺畅。
- **[2] B**：
  - **线索**：解释酵母使面团发酵的原理。空前描述"add yeast to flour and water, something amazing happens"，空后详述"The yeast is a tiny living organism...produces carbon dioxide gas...forms bubbles in the dough, making it rise"。B选项"That is because the yeast causes a chemical reaction that changes the structure of the dough"（那是因为酵母引发化学反应，改变面团结构）中"That is because"解释空前"something amazing"的原因，"chemical reaction...changes the structure"与空后酵母产气、面团膨胀的机制精准对应，衔接紧密。
- **[3] C**：
  - **线索**：食物受热变化的科学规律。空前提问"Why does an egg turn solid when you heat it?"，空后解释"The protein molecules...unfold and connect...changes the liquid egg into a solid...similar to...jelly"。C选项"In fact, the way food changes when we cook it follows clear scientific rules"（事实上，食物在烹饪时的变化遵循明确的科学规律）中"In fact"引出科学解释，"way food changes...scientific rules"与空后蛋白质分子变化的机制呼应，衔接自然。
- **[4] D**：
  - **线索**：烹饪技巧与化学知识的运用。空前提出"Some cooking tricks are actually chemistry lessons"，空后列举盐加速意面、腌肉使肉更嫩、切菜方式影响风味等实例。D选项"These are practical examples of how knowing a little chemistry can improve your cooking"（这些是了解一点化学如何改善烹饪的实用例子）中"These"指代空前"cooking tricks"，"practical examples"总领空后多个实例，"improve your cooking"与空后各技巧的效果对应，衔接得当。

- [5] E：
    - **线索**：全文收束与升华。空[5]位于结尾段段首，空后总结"Cooking and chemistry go hand in hand...kitchen is like a lab...experiment, learn, and create...think about the science behind your food"。E选项"In short, every meal you make is a chance to learn something new about science"（简言之，你做的每顿饭都是学习科学新知识的机会）用"In short"总结全文，"learn something new about science"与空后"experiment, learn"及"think about the science"完美呼应，收束精准。
- **干扰项分析**：
    - **F. For instance, professional chefs often study food science to improve their skills.**：涉及专业厨师学习食品科学，易被误填[4]。但F以"For instance"引出职业厨师这一特定群体，而空[4]前后文讨论的是普通人可用的烹饪技巧（加盐、腌制、切菜），空后列举的是家庭厨房中的实用方法，并非专业厨师的学习行为；F的"professional chefs"与空后"pinch of salt""marinating""cut vegetables"等日常技巧层级不匹配。F属"主题相关但衔接对象错位"型干扰。